

Table 1: A comparison with other blockchain frameworks

Feature	Hyperledger Fabric	Ethereum	Bitcoin
Network type	Permissioned	Permissionless	Permissionless
Privacy and confidentiality	Supports private transactions and confidentiality	Public ledger	Public ledger
Smart contracts	Chaincode (supports multiple languages)	Solidity (specific to Ethereum)	Bitcoin Script (limited scripting language)
Consensus mechanism	Pluggable consensus (e.g., Raft, Kafka)	Proof of Work (PoW)	Proof of Work (PoW)
Scalability	Modular architecture allows for scalability	Challenges with scalability	Challenges with scalability
Identity management	Provides robust identity management capabilities	Limited identity management capabilities	Limited identity management capabilities
Customisation	Highly customisable	Less customisable	Less customisable
Use cases	Enterprise applications, supply chain management, identity management	Decentralised applications, tokenisation, decentralised finance	Digital currency, peer-to-peer transactions

are using Hyperledger Fabric to create a unified and secure patient database, enabling authorised healthcare professionals to access patient information quickly and accurately, leading to improved patient care and outcomes.

- Ensures data privacy and integrity
 - Enables quick and accurate access to patient information
 - Streamlines healthcare processes and improves efficiency
3. **Finance and banking:** Hyperledger Fabric is reshaping the finance and banking sector by providing a secure and transparent platform for financial transactions. For instance, in cross-border payments, Hyperledger Fabric is used to facilitate faster and more cost-effective transactions between banks and financial institutions. Additionally, Hyperledger Fabric is used for trade finance, enabling secure and efficient transactions between buyers and sellers across the globe. This has the potential to revolutionise international trade by reducing the time and cost associated with traditional trade finance processes.
- Facilitates faster and cost-effective transactions
 - Enables secure and efficient trade finance processes
 - Enhances transparency and reduces fraud in financial transactions
4. **Government:** Governments are leveraging Hyperledger Fabric for various applications, including identity management and public service delivery. For instance, in identity management, governments use Hyperledger Fabric to create secure and tamper-proof digital identities for citizens. This helps in reducing identity theft and fraud, while also improving access to government services. Moreover, Hyperledger Fabric is being used for land registry, enabling governments to create a transparent and secure system for recording land ownership, leading to more efficient land management and reduced disputes.

- Reduces identity theft and fraud
- Improves access to government services
- Enhances transparency and efficiency in government operations

1. **Honeywell**

Industry: Diversified materials and products

Overview: Honeywell implemented Hyperledger Fabric to enhance the traceability and transparency of its supply chain for materials and products. By utilising blockchain technology, Honeywell can track the origin of raw materials, monitor manufacturing processes, and ensure the authenticity of its products.

Results: The use of Hyperledger Fabric has enabled Honeywell to improve supply chain efficiency, reduce counterfeit products, and enhance trust among customers and stakeholders.

2. **Bosch**

Industry: Automobiles and auto parts

Overview: Bosch adopted Hyperledger Fabric to streamline its supply chain processes for automobile components. By digitising the supply chain, Bosch can track the movement of parts from suppliers to assembly lines, improve inventory management, and enhance quality control.

Results: The implementation of Hyperledger Fabric has enabled Bosch to reduce lead times, optimise inventory levels, and improve the overall efficiency of its supply chain operations.

3. **NP Solution Limited**

Industry: Human resources and staffing